

SELENE-MATCH

Multi-modal, Sun-Angle and Scale-Invariant Lunar Image Correspondence
Chandrayaan-2 (OHRC · TMC-2 · IIRS) ↔ LRO NAC Reference

PS ID	26166
Organisation	ISRO / Department of Space
Category	Software — Space Technology
This document	Consolidated final blueprint, reconciling two independent AI-generated drafts (Grok-style playbook & GLM-style blueprint) against real Chandrayaan-2 payload specs
Team size	4 members — role split included
Version	Final — August 2026

0 · How the Two Drafts Were Reconciled

You had two independent AI drafts for PS 26166: **Draft A** (the "playbook" — geometry-first, DEM/hillshade-based, ensemble deep-learning matchers, ISIS/ASP-aware) and **Draft B** (the "blueprint" — phase-congruency descriptor, Lambertian correction, a 3-DOF selenographic-sphere RANSAC). Both are usable, but they disagree on facts and on the core illumination-invariance strategy. This document keeps what is correct and buildable in each, discards what is wrong or unrealistic for a 4-person team, and gives you one architecture to build — not two to choose between at 2 a.m.

0.1 · Fact-check: the two drafts disagree on sensor specs

Parameter	Draft A (playbook)	Draft B (blueprint)	Verdict
OHRC GSD	0.25 m	0.25 m	Agree — correct.
OHRC swath	~3 km	~10 km	Draft A correct. ISRO payload notes cite ~3 km swath for OHRC's high-res strip.
TMC-2 GSD / swath	5 m / ~20 km	5 m / ~20 km	Agree — correct.
IIRS GSD	80 m	~20 m	Draft A correct. Published IIRS-to-WAC registration literature and ISRO specs consistently cite ~80 m spatial sampling, not 20 m. This is a big error in Draft B — it understates the real scale ratio by 4×.
OHRC:IIRS scale ratio	~320×	~80×	Draft A correct , as a consequence of the GSD error above. Design for 320×, not 80×.
References cited	NASA ASP Chandrayaan-2 cookbook, USGS ISIS, a 2025 IIRS→WAC SIFT paper (~73 m RMS), CNSFM 2025, LightGlue/RoMa/MINIMA papers	Generic mentions of ISIS3/AutoNet/SPICE, no dataset-specific literature RMSE baseline	Draft A is better grounded — it gives you a real published number (~73 m RMS, IIRS mid-latitude) to beat in your PPT.
Self-scoring	None	Gives itself "9.6/10 expected SIH score"	Ignore this. A draft scoring itself is not a jury signal, and is a classic overconfidence tell — treat it as a red flag, not a plus.

Action: Memorise Draft A's numbers for your PPT and code (GSD, swath, scale ratios). Do not repeat Draft B's IIRS GSD or OHRC swath figures to judges — an ISRO evaluator will catch it immediately.

0.2 · What each draft got right — keep it

Idea	Source	Verdict	Why
Map-project first, match second (never raw-pixel homography)	Draft A	KEEP	Pushbroom + sphere + terrain means a single homography cannot work on a real strip. This is the single most important architectural decision in either draft.
Gated ensemble of matchers by illumination-delta / modality / scale, not one algorithm for everything	Draft A	KEEP	No single method (classical or deep) handles same-sun OHRC↔NAC and opposite-sun, cross-modal IIRS↔WAC equally well. A gate is honest engineering, and it is a great slide.
Phase congruency as an illumination-invariant classical descriptor	Draft B	KEEP, DEMOTE	Real, well-published signal-processing technique (Kovesi). Genuinely brightness/contrast invariant. But it is <i>not proven against true polarity reversal</i> (crater rim bright→dark flip) the way hillshade/crater-graph matching is. Use it as a fast, DEM-free fallback expert in the gate — not the sole illumination strategy.
Selenographic (spherical, 3-DOF rotation) geometric model as an alternative to planar homography	Draft B	KEEP, SCOPE DOWN	Correct physics reasoning (Moon R=1737.4 km, so planar homography is wrong over multi-km footprints) and cheaper to implement than full ISIS/SPICE if you are short on prep time. Use it as your fallback geometric backend when a DEM/ISIS pipeline isn't available; use DEM map-projection + piecewise/TPS as the primary path if P1 can get it working.
Hillshade / crater-neighbourhood-graph matching for large sun-angle deltas	Draft A	KEEP	This is the only approach in either draft that actually survives a documented sun-azimuth flip, because it matches topology, not shading.
Inverse-compositional Lucas-Kanade for sub-pixel refinement	Both	KEEP	Both drafts converge here independently — strong signal it's the right call. O(N) per iteration, standard, well-understood, ISRO-legible.
Grid/quadtrees-based uniform-distribution sampling of match points, with an explicit uniformity/coverage score	Both	KEEP	The PS explicitly asks for this and neither classical SIFT nor raw deep matchers give it to you for free.
Full ISIS3 + SPICE + Ames Stereo Pipeline install as the primary geometry backend	Draft A (optional)	OPTIONAL ONLY	Huge credibility if it works, but it is a multi-day install with kernel downloads that can run to hundreds of GB. For a 4-person team, treat as a stretch goal behind a feature flag, not a dependency of the core pipeline.

Six-person team split, Docker/K8s, Dash *and* React, dense matcher (RoMa) as a default

Both (over-scoped)

DROP

You have 4 people, not 6, and a laptop, not a GPU cluster necessarily. Cut to what \$6-\$9 below specify.

1 · What ISRO Is Actually Asking

Image registration aligns a **source** (moving) Chandrayaan-2 image — OHRC, TMC-2 or IIRS — to a **reference** (fixed) lunar control image, explicitly LRO NAC, with implied WAC/DEM control for coarser sensors. The deliverable is not "an AI that finds craters." It is generic software that outputs correspondence, a registered product, and evaluation metrics, at sub-pixel accuracy on the source image's own grid, with match points uniformly distributed.

1.1 · The three challenges as engineering constraints

Challenge	What actually breaks	What you must build
Illumination / sun angle Different orbits → different sun azimuth & elevation. No atmosphere → shadows are binary. Opposite azimuth reverses crater polarity (bright rim ↔ dark rim).	SIFT/ORB/AKAZE/SuperPoint (gradient- or learned-shading-based), NCC on raw DN, CLAHE-as-a-fix.	Hillshade/DEM relight matching, crater-topology graph matching, and/or phase-congruency descriptors — gated by measured Δsun-azimuth.
Viewpoint Different orbit tracks, pushbroom linescan (not a pinhole frame), terrain relief on a 1737 km sphere.	A single global homography (<code>cv2.findHomography</code> on the whole strip).	Map-project (or selenographic sphere model) onto a common frame first; residual 2D field as piecewise-affine / TPS, not one 8-DOF matrix.
Scale OHRC 0.25 m, TMC-2 5 m (20×), IIRS 80 m (320× vs OHRC). NAC ~0.5 m, WAC ~100 m.	SIFT octaves (~8–16× typical), feeding full-res OHRC into a 1024-px matcher.	Explicit GSD pyramid in <i>metres</i> , always matching at a common metres/pixel level, then refining downward to source GSD.

1.2 · Expected-solution checklist (treat as your acceptance test)

- Registered product (source warped into reference frame) + match points, in pixel *and* ground coordinates.
- Metrics: RMSE (px and m), inlier count, inlier ratio, plus a uniformity/coverage score (implied, not named — you must define and report it).
- One generic software path for OHRC, TMC-2 and IIRS, with modality-specific branches — not three disconnected scripts.
- Sub-pixel accuracy on the *source* image's native GSD, not "looks aligned at display zoom."
- Uniform spatial distribution of matches — no clustering on a single crater rim.
- Honest failure handling: shadow voids, high-latitude IIRS, 320× scale gaps shown as a mask, not hidden.

1.3 · Sensor cheat-sheet (memorise before judging)

	OHRC	TMC-2	IIRS	LRO NAC (ref)	LRO WAC (ref)
Role	Highest-res PAN	Stereo mapper	Hyperspectral IR	Fine geometric control	Coarse geometric control
Band	PAN 0.4–0.85 μm	PAN, 3-line stereo ±25°	0.8–5.0 μm, ~256 bands	PAN	Visible, wide
GSD @100km	0.25 m	5 m	80 m	~0.5 m	~100 m
Swath	~3 km	~20 km	~20 km	~5 km	global mosaic
Geometry	Pushbroom, steerable	3-line pushbroom	Pushbroom cube (QUB)	Pushbroom	Pushbroom mosaic
Hard part	Size, shadows, jitter	Illum vs OHRC, 20× scale	Modality + 16–320× scale, high-lat	Different sun than CH2	Low res

Memorise the ratios: OHRC:TMC ≈ 20:1, TMC:IIRS ≈ 16:1, OHRC:IIRS ≈ 320:1. IIRS is a spectrometer, not "another camera" — you will pick 1–3 high-SNR bands (or a PCA component) and treat that as the moving image; never feed 256 bands into a matcher. A published SIFT-to-WAC study on IIRS reports mean RMS ~73 m at mid-latitude, failing past ±55° latitude — that is your baseline to beat and your known failure region to show honestly.

2 · Similar Software — What Exists, What It Solves, Why You Can't Just Wrap It

ISRO did not ask you to reinvent planetary photogrammetry. Existing tools are your backends, baselines and "related work" slide — never your product. If a judge suspects you just wrapped Ames ASP, you lose the round.

2.1 · Planetary photogrammetry (closest cousins)

Software	What it solves	Why it isn't your product
USGS ISIS3	PDS ingest, SPICE camera init, map-projection, control networks, bundle adjustment — the planetary-imaging standard, includes Chandrayaan-2 camera models.	Steep CLI, huge SPICE kernel downloads, matching inside it is classical and illumination-fragile, not a correspondence UI.
NASA Ames Stereo Pipeline (ASP)	Stereo DEMs/orthos from CH2 OHRC/TMC-2, with a documented Chandrayaan-2 cookbook (<code>bundle_adjust</code> , <code>parallel_stereo</code> , <code>mapproject</code> , <code>pc_align</code>).	Assumes stereo pairs with <i>similar</i> illumination; not multi-modal (no IIRS path); not a generic matcher.
USGS AutoNet	Automated control-network generation, SIFT-based, tuned for same-sensor (LRO NAC↔NAC) registration.	Not cross-mission, limited illumination handling, not open-source.
SOCET SET / BAE Systems	Commercial gold-standard planetary mapping, historically used by NASA/USGS.	Closed-source, licensed — mention only as an industry baseline.

2.2 · Earth remote-sensing registration (steal ideas, not the product)

Software	What it solves	Why it isn't your product
COSI-Corr (Caltech)	Sub-pixel dense frequency-domain correlation for co-seismic/glacier displacement on optical satellite pairs.	Same-modality Earth orthos only; no sun-azimuth or crater logic. Steal the sub-pixel phase-correlator idea.
MicMac (IGN France)	Open photogrammetry, dense matching.	Earth-centric, no PDS/SPICE, no lunar illumination model.
AROSICS / GDAL co-registration	Local phase-correlation co-registration and RPC-based warping for Earth satellites.	Excellent warp engine (you will use GDAL directly) but not multi-modal IR, not polarity-aware.
ENVI / QGIS Georeferencer	Manual GCP placement + polynomial/TPS warp — what planetary scientists do today for IIRS.	Manual, slow. Your automated pipeline must beat this on time and match the ~1-IIRS-pixel literature accuracy.

2.3 · Academic matchers — your ensemble's ingredients

Method	Solves	Fails when	Role in SELENE-MATCH
SIFT / AKAZE	Scale, modest rotation/illumination	Sun-azimuth flip, IR vs PAN, huge scale	Baseline for comparison slide; IIRS→WAC classical fallback
Phase congruency descriptor	Brightness/contrast invariance (frequency-domain)	True polarity reversal not fully solved; slower than sparse DL	DEM-free illumination-robust fallback expert
SuperPoint / ALIKED + LightGlue	Fast, strong learned sparse matching	Polarity reversal, extreme cross-modality	Default expert for similar-sun, similar-modality tiles
RoMa / dense matchers (optional, GPU)	Dense correspondence, low-texture mare	Slow, VRAM-hungry	Optional expert on hard tiles if a GPU is available
Mutual Information (SimpleITK)	Classic multi-modal registration	Needs a decent initial alignment	IIRS ↔ WAC/TMC branch
Phase correlation / Fourier-Mellin	Translation (+scale/rotation) prior	Repetitive craters, large relief	Coarse stage after map-projection, on every pair
Crater-neighbourhood graph matching	Multi-illumination, polarity-robust	Needs a working crater detector; weak on smooth mare	Sun-angle backbone when Δazimuth is large

Positioning line for your PPT: "ISIS/ASP give us camera models when available; COSI-Corr gives us the sub-pixel correlator pattern; LightGlue gives us modern appearance matching; phase congruency and crater graphs give us sun-angle invariance. SELENE-MATCH is the software that chooses among them per pair, enforces uniform ground-control points, and emits ISRO-usable products — something none of the above does end-to-end, in the open, across all three Chandrayaan-2 payloads."

3 · Final Locked Tech Stack

One conda environment. GDAL and PyTorch are the two packages that eat a weekend if installed ad hoc — get GDAL/rasterio from `conda-forge`, and torch/kornia/lightglue from pip inside that env, on day one, before any code is written.

3.1 · Core science stack (must-have)

Package	Role
Python 3.11, numpy, scipy	Arrays, FFTs (phase congruency, phase correlation), peak fitting
opencv-python-headless	SIFT/AKAZE baseline, warping, MAGSAC++ (<code>USAC_MAGSAC</code> flag)
scikit-image	Phase correlation, corner peaks, morphology for shadow masks
rasterio, GDAL, pyproj, shapely, affine	GeoTIFF I/O, warping, CRS handling (Moon equirectangular / stereographic)
pvl / planetaryimage	PDS3/PDS4 label and QUB parsing for CH2 products
pydantic, pyyaml, rich, loguru	Config, CLI, structured logs
pytest	Synthetic-transform unit tests (know-your-recovery-error before touching real data)

3.2 · Matching & illumination stack (must-have)

Package	Role
PyTorch 2.x (CUDA if available, CPU fallback mandatory)	Runtime for LightGlue / optional dense matcher
<code>lightglue</code> (cvg/LightGlue) + ALIKED/SuperPoint	Default sparse matcher for similar-sun, similar-modality tiles
Custom phase-congruency module (log-Gabor bank, own implementation)	Illumination-invariant descriptor; DEM-free fallback
SimpleITK	Mattes mutual-information registration for the IIRS branch
Custom hillshade + crater-graph module	Sun-angle backbone for large Δ azimuth pairs (needs a DEM clip)
OpenCV <code>USAC_MAGSAC</code>	Robust outlier rejection (do not use vanilla RANSAC as the last word)

3.3 · Geometry backend (tiered, so the team never blocks on install pain)

Tier	Approach	When to use
Tier 1 (stretch)	USGS ISIS3 + NAIF SPICE kernels + optional Ames ASP <code>mapproject</code>	If P1 gets it installed and working with time to spare — big credibility win, cite in PPT even if used on only one pair
Tier 2 (default)	PDS-label footprint + sun angles → affine/RPC initial map-projection onto a WAC/NAC mosaic clip; DEM clip from SLDEM2015/LOLA where available for hillshade	Default path for all 4 team members — must work with zero manual installs beyond the conda env
Tier 3 (fallback geometric model)	Selenographic sphere model: project matched pixels onto a sphere of radius 1737.4 km, solve a 3-DOF rotation (Kabsch/SVD) instead of an 8-DOF homography	Whenever a DEM isn't available for a pair, or a tile is far from nadir — cheaper than Tier 1, geometrically more honest than a bare homography

3.4 · Product / UI stack (must-have, kept intentionally small)

Package	Role
FastAPI + uvicorn	Job API: submit pair → run pipeline → poll status → fetch products
React + TypeScript + Vite (single page)	Workbench: pair picker, run log, match overlay, scoreboard, export button
Canvas/SVG overlay (no heavy gigapixel viewer needed for 4 people)	Draw match lines colour-coded by expert, residual arrows, checkerboard wipe on the precomputed showcase pairs
plotly / recharts	RMSE, inlier ratio, coverage-grid heatmap
reportlab / matplotlib	Auto-generated one-page PDF report per job

3.5 · Explicitly out of scope for a 4-person team

Kubernetes, Kafka, MLflow, multi-agent orchestration · CycleGAN/diffusion relighting as a deliverable (research-grade, not 36 h) · COLMAP/SfM (wrong geometry class — you already have orbiter geometry, not a drone survey) · training a foundation model from

scratch on Chandrayaan data · a second UI framework (pick React *or* Dash, not both) · RoMa/dense matching as the *default* path (GPU-only, keep it optional).

4 · System Architecture

Ingest (PDS/GeoTIFF + metadata + sun angles) → Geometry (map-project / selenographic model, GSD pyramid) → Illumination normalise (hillshade / phase congruency / census) → Gated matcher ensemble → MAGSAC++ + uniform GCP sampling → Warp (piecewise/TPS) → Sub-pixel refine (IC-LK) → Product export + Evaluation dashboard

Matching is deliberately not the first stage. Geometry and illumination normalisation do more for accuracy than any single matcher, and separating the stages is what lets four people work in parallel without stepping on each other's code.

4.1 · Layer contracts (how the team splits work without merge conflicts)

Layer	Input	Output	Rule
Ingest	Folder of PDS4/GeoTIFF files	A canonical <code>Pair</code> object: arrays or COG paths, GSD, sun az/el, footprint, CRS, modality tag	No one reads a raw file except this layer
Geometry	Canonical pair + DEM (optional)	Both images on a common grid, plus a transform back to source pixels	Matching never runs in raw, unmatched camera geometry
Match	Common-grid tiles + Δ sun + modality tag	Inlier matches in grid px, with a per-match method-id	Every match knows which expert produced it — needed for the "method mix" slide
Product	Matches + source/reference rasters	Warped GeoTIFF, matches.csv, metrics.json, residual plot	Source-pixel RMSE computed only after mapping back to source geometry
Workbench	Job id	Overlay, wipe, scoreboard, ZIP download	UI never reimplements matching logic

5 · Algorithm Pipeline — Stage by Stage

Stage 0 — Pair validity screen

Compute footprint overlap, GSD ratio, Δ sun-azimuth, Δ sun-elevation, and the modality pair before matching. Refuse (with a stated reason) if overlap <15%. Flag polarity risk if $|\Delta$ azimuth| >90°. Flag mandatory pyramiding if GSD ratio >32. This screen alone signals domain understanding to a judge before a single match point exists.

Stage 1 — Ingest and radiometry

- Read PDS4 (IMG+XML or QUB) or GeoTIFF; keep native DN, don't crush to 8-bit until a matcher demands it.
- Parse acquisition time, sun azimuth/elevation, incidence angle, footprint, instrument id.
- For IIRS: pick 1-3 high-SNR bands (e.g. near 1.2-1.6 μ m) or one PCA component as the optical proxy — never feed all 256 bands to a matcher.
- Build a shadow mask (robust low-DN percentile); no downstream matcher may vote inside it.

Stage 2 — Geometry: get onto a common grid

Preferred (Tier 1): ISIS `spiceinit` → `mapproject` both images onto the same DEM. Default (Tier 2): use the footprint + sun metadata to crop the reference mosaic and estimate an initial affine. Fallback (Tier 3): selenographic sphere model. Build a pyramid *in metres* — e.g. 80 m (IIRS/WAC), 20 m, 5 m (TMC), 1 m, 0.25 m (OHRC) — and always start matching at the coarsest common level.

Stage 3 — Illumination normalisation (gated)

Condition	Technique
$ \Delta$ azimuth < 30°	Rank/census transform or high-pass (Lambertian-style DN correction); enough for LightGlue directly
$ \Delta$ azimuth large, DEM available	Hillshade the DEM at each image's own sun position; match image↔own-hillshade, then hillshade↔hillshade
$ \Delta$ azimuth large, no DEM	Phase-congruency descriptor on both raw images (frequency-domain, brightness/contrast invariant)
DEM poor / detector fails	Fall back to crater-neighbourhood graph matching — do not let CLAHE stand in as "the fix"

Stage 4 — Coarse correspondence

(a) Phase correlation (FFT) on coarsest overlapping tiles for a translation prior. (b) Crater-neighbourhood graph: detect crater centres/radii on both images, build a k-NN angle/distance graph, match graphs — 8-20 good crater pairs already constrain an affine on a TMC tile, and this path is illumination-invariant by construction. Skip gracefully on smooth mare with no craters.

Stage 5 — Fine appearance matching (the gate table — put this in the PPT)

Gate condition	Expert used
Same modality, $ \Delta$ az <40°, GSD ratio <4	ALIKED/SuperPoint + LightGlue on 1024-1600 px tiles
Same modality, low texture (mare)	Dense matcher on hard tiles if GPU available, else phase congruency
$ \Delta$ az >60° (polarity risk)	Do not run appearance matchers on raw DN — hillshade or crater-graph only
IIRS ↔ WAC/TMC	Mutual information + SIFT-on-stretched-band baseline (beat the ~73 m published RMS)
OHRC ↔ TMC (20× scale)	Match at the coarser common GSD first, propagate down — never match 0.25 m to 5 m natively

Stage 6 — Outlier rejection and uniform GCP sampling

Fit a residual affine/quadratic with MAGSAC++ (reprojection threshold in metres, not mixed-GSD pixels). Drop matches inside the shadow mask or below a score threshold. Partition the overlap into an 8×8 grid; keep the best 2-5 inliers per cell with a minimum spacing constraint. If a cell is empty, retry locally before giving up; mark truly empty cells as "no-texture/shadow," not as pipeline failure — this is exactly how you satisfy "uniform distribution."

Stage 7 — Warp

Estimate a thin-plate spline or piecewise affine from the uniform GCPs on the common grid; warp with cubic/Lanczos via GDAL; nodata in shadows; write a Cloud-Optimised GeoTIFF, plus the inverse transform for source-pixel residual reporting.

Stage 8 — Sub-pixel refinement

For each kept GCP, run inverse-compositional Lucas-Kanade on a 21-41 px patch (or a phase-correlation peak fit) to refine the match. Typical gain: 0.3-1.0 px. Report RMSE both before and after this stage — that before/after plot is one of your strongest slides.

6 · Geometric Models & Uniformity Metric

6.1 · Which 2D model, when

Model	Use when	Do not use when
Translation	Post map-projection, tiny residual (FFT stage)	As a final OHRC-strip model
Similarity / affine	Single TMC/IIRS tile, good crater GCPs	A full OHRC strip
Homography	A small, planar-ish tile (<2 km, mare)	Any long strip or highland terrain
Selenographic 3-DOF rotation	No DEM available, or a tile far from nadir	Once you already have DEM map-projection working — redundant then
Piecewise affine / TPS	Default final residual model	Fewer than ~12 well-spread GCPs

6.2 · Sub-pixel methods (implement at least one, ideally two)

- **Spatial:** inverse-compositional Lucas-Kanade on illumination-normalised patches.
- **Frequency:** phase-correlation peak interpolation (parabolic/sinc fit) — strong on textured orthos, borrowed from COSI-Corr.
- Report RMSE before and after refinement using a *held-out* set of GCPs, never the same points used to fit the warp.

6.3 · Uniformity / coverage metric (define this explicitly — the PS implies it but doesn't name it)

Divide the overlap into an $n \times n$ grid ($n=8$). Let $c_i = 1$ if cell i has ≥ 1 inlier. **Coverage** = $\text{mean}(c_i)$, excluding shadow-only cells from the denominator. Combine with a normalised nearest-neighbour spacing term for a single **Uniformity score U**. Targets: Coverage ≥ 0.7 on TMC/NAC, ≥ 0.6 on OHRC, ≥ 0.5 on IIRS.

7 · Evaluation Protocol

Metric	Definition	Why it matters
N_raw / N_inlier	Matches before / after MAGSAC++	Shows each expert is alive; PS asks for inlier count
Inlier ratio	N_inlier / N_raw	PS explicitly asks for this
RMSE_px / RMSE_m	RMS residual on held-out GCPs, in source pixels and metres	PS's core sub-pixel accuracy requirement
CE90 / P90	90th-percentile residual (m)	ISRO-style accuracy language, robust to one bad point
Coverage / Uniformity U	Grid occupancy + spacing, shadow-aware	PS's "uniform distribution" requirement
Method mix	% inliers per expert (crater graph / LightGlue / phase-congruency / MI)	Proves the ensemble is real, not a sticker on one algorithm

Honest targets for the PPT: OHRC↔NAC (similar sun): RMSE 0.3–0.8 OHRC px, inlier ratio >0.4, coverage >0.6. TMC↔NAC: RMSE <0.5 TMC px. IIRS↔WAC: RMSE <1 IIRS px at mid-latitude (beats the published ~73 m baseline). Opposite-azimuth pair: show your crater/hillshade path succeeding while a plain SIFT baseline's inlier ratio collapses — this comparison is your single best slide. Never claim sub-0.05-pixel accuracy without a held-out check; judges will ask how you validated it.

8 · Repository Layout & Deliverables

Path	Contents
src/selene/ingest/	PDS4/GeoTIFF readers, sun/footprint parsing
src/selene/geometry/	CRS, map-projection wrapper, selenographic model, GSD pyramid
src/selene/illum/	Hillshade, phase congruency, census transform
src/selene/craters/	Crater detection, neighbourhood graph, graph matching
src/selene/matchers/	LightGlue, phase-correlation, mutual-information (SITK), SIFT baseline
src/selene/robust/	MAGSAC++ wrapper, uniform GCP sampler
src/selene/warp/	TPS/piecewise warp, GDAL export, IC-LK sub-pixel refine
src/selene/eval/	Metrics, residual plots, uniformity score
api/	FastAPI job endpoints
ui/	Vite React workbench
data/samples/	4 pre-cleared demo pairs: OHRC↔NAC, TMC↔NAC, IIRS↔WAC, one opposite-azimuth pair

8.1 · Export bundle (the "registered product")

- `registered.tif` — warped source in the reference CRS, nodata in shadows
- `matches.csv` — id, source/ref pixel & lon/lat, residual (px, m), method, inlier flag, score
- `metrics.json` — the full Section 7 scoreboard
- `residual.png`, `checkerboard.png` — visual QA
- `report.pdf` — auto-generated one-pager per job

9 · Four-Person Team — Role Split & Interfaces

Four roles, mapped directly onto the layer contracts in §4.1, so no two people edit the same module. Each role owns one git-visible directory and one CLI/API contract with the others.

Role	Owns	Builds	Deliverable at freeze
P1 — Geometry & Ingest Lead	<code>ingest/</code> , <code>geometry/</code>	PDS4/GeoTIFF/QUB readers · sun angle & footprint parsing · GSD pyramid builder · Tier 2 affine map-projection · (stretch) Tier 1 ISIS/SPICE wrapper · Tier 3 selenographic sphere model	A <code>Pair</code> object for any sample file, on a common metres-grid, in <5 s
P2 — Illumination & Structural Matching	<code>illum/</code> , <code>craters/</code>	Hillshade generator from DEM clips · phase-congruency descriptor (log-Gabor bank) · crater detector + neighbourhood-graph matcher · the Stage-3 gating logic	Working image↔hillshade and phase-congruency matching on the opposite-azimuth demo pair
P3 — Appearance Matching, Robust Fitting & Sub-pixel	<code>matchers/</code> , <code>robust/</code> , sub-pixel refine in <code>warp/</code>	LightGlue integration · SIFT baseline for the comparison slide · mutual-information IIRS branch · MAGSAC++ wrapper · uniform GCP sampler · inverse-compositional LK sub-pixel refinement	End-to-end matches→inliers→uniform-GCPs→sub-pixel pipeline on OHRC↔NAC and TMC↔NAC
P4 — Product, Evaluation & Workbench	<code>eval/</code> , <code>warp/</code> (GDAL export), <code>api/</code> , <code>ui/</code>	TPS/piecewise warp + GeoTIFF export · metrics.json + residual/checkerboard plots · uniformity score · FastAPI job endpoints · React workbench (pair desk, run log, overlay, scoreboard, export) · PPT + demo script	A judge can pick a sample pair in the browser, run it, and download the full export bundle

Daily sync rule: the four `Pair / Match / Product` object schemas in §4.1 are frozen after Phase A (below). Nobody changes a field name without a 2-minute stand-up with the other three — this is what lets four people integrate in hours, not days, during the finale.

10 · Roadmap — Now → Internal Round → Grand Finale

10.1 · Phase A — This week: problem understanding & data

1. Register on PRADAN/MapBrowse; pull 2 OHRC, 2 TMC-2 (nadir), 2 IIRS cubes, overlapping LRO NAC frames, one WAC mosaic clip, one DEM clip.
2. Open everything in QGIS; manually drop 8 GCPs on one IIRS↔WAC pair and one OHRC↔NAC pair — this is your human accuracy baseline.
3. Every member can print sun az/el and GSD from a label without looking it up.
4. P1 freezes the `Pair` object schema; everyone writes against it from day one.

10.2 · Phase B — Two weeks: geometry & classical baseline

1. P1: map-projection (Tier 2 default) onto a common grid; visually verify overlay in QGIS.
2. P3: phase correlation + SIFT + MAGSAC + GDAL warp — this is "Baseline A," needed for the comparison slide.
3. P4: eval CLI with RMSE/inliers; run Baseline A on a same-sun pair (should look decent) and an opposite-sun pair (should look bad) — save both numbers, this is your "why we exist" slide.
4. P2: shadow mask + DEM hillshade generator.

10.3 · Phase C — Two weeks: the winning physics

1. P2: hillshade-to-hillshade matching; crater detector + neighbourhood graph (even a brittle version wins the opposite-azimuth pair); phase-congruency descriptor as the DEM-free fallback.
2. P3: LightGlue on tiles, wired into the Stage-5 gate table as code, not a comment; mutual-information IIRS branch benchmarked against the ~73 m literature figure.
3. P3/P4: uniform GCP sampler + TPS/piecewise warp + IC-LK sub-pixel; before/after RMSE plot.
4. P1 (stretch): attempt Tier 1 ISIS/SPICE install in a Linux VM; keep Tier 2/3 as the tested fallback regardless.

10.4 · Phase D — Internal college round

1. Workbench with 2–3 precomputed jobs and one live fast-path pair (TMC, smallest data).
2. 10-slide deck (see §11); a short clip of SIFT failing vs your method succeeding on the opposite-azimuth pair.
3. Never promise "we'll add X later" — show only what runs today.

10.5 · Phase E — Between internal and finale

1. Harden ingest for whatever the official TBD dataset link turns out to be — do not hard-code file names or one folder layout.
2. Optional: dense matcher on hard tiles if a GPU is available; ISIS Tier 1 backend if it's working, for extra credibility.
3. Auto-generated report.pdf; a synthetic polarity-flip unit test (invert + sun-aware DEM render) that shows crater-graph matching still works while SIFT collapses — automate this as a regression test, not a manual demo.

10.6 · Phase F — 36-hour grand finale clock

Hours	Task	Owner
0–4	Ingest the official dataset; produce <code>Pair</code> objects. Adapt parsers if labels differ — do not retune matchers yet.	P1 + P4
4–12	Run the full pipeline on official pairs; collect metrics; if a pair fails, switch gate (hillshade vs crater vs MI), not the whole architecture.	P2 + P3
12–20	Precompute showcase jobs; get the smallest official pair running live in the UI; polish residual plots and export ZIP.	P3 + P4
20–28	Feature freeze. PPT numbers pulled from official runs; record a backup demo video; run one ablation (SIFT vs SELENE-MATCH on the hardest pair).	All
28–36	Sleep in shifts; two dry-runs of the 3-minute demo; print the metric table on paper as a backup.	All

11 · Demo, PPT & Judging

11.1 · Ten slides

1. Title — SELENE-MATCH, PS 26166, one sentence: "sub-pixel correspondence when the Sun moves and the camera changes."
2. Problem as physics — the same crater at two suns and two GSDs, polarity reversal annotated.
3. Why not SIFT — your Baseline A numbers, opposite-azimuth collapse.
4. Architecture — the §4 pipeline diagram; metadata first, ensemble gated.
5. Illumination trick — hillshade + crater graph + phase congruency, one diagram.
6. Scale trick — the metres-based GSD pyramid; OHRC never matched to IIRS natively.
7. IIRS branch — band selection + mutual information; literature 73 m vs your number.
8. Results — the §7 scoreboard on 3+ pairs; checkerboard; residual heatmap.
9. Product — workbench screenshot, export bundle, CLI; related work positioned as backends, not competitors.
10. Ask / roadmap — generic, reproducible, ready for the official TBD dataset; team roles.

11.2 · Three-minute live demo script

1. 0:00-0:20 — Open the pair desk: OHRC vs NAC with near-opposite sun vectors. "This is the case that breaks SIFT."
2. 0:20-0:50 — Play the precomputed SIFT overlay (sparse, clustered, wrong) then yours (spread, colour-coded by method).
3. 0:50-1:20 — Checkerboard wipe at full zoom; quote RMSE_px live.
4. 1:20-1:50 — Switch to IIRS vs WAC: "different physics, same software, different gate."
5. 1:50-2:20 — Scoreboard: inliers, ratio, RMSE, coverage; click an empty shadow cell on purpose.
6. 2:20-2:45 — Export ZIP, open matches.csv in the viewer.
7. 2:45-3:00 — One sentence on sub-pixel LK before/after.

11.3 · Hard questions, rehearsed answers

Question	Answer
Why not just SuperGlue/LightGlue for everything?	Appearance matchers still encode shading. Opposite azimuth reverses crater polarity; we gate to hillshade/crater-graph/phase-congruency matching when Δ azimuth is large, and reserve LightGlue for similar-sun pairs.
Is this photogrammetry or computer vision?	Both. Geometry (map-projection or the selenographic sphere model) gets us onto a common grid; CV supplies residual correspondence; photogrammetric warping writes the product.
How is it actually sub-pixel?	Inverse-compositional Lucas-Kanade / phase-peak fitting on patches after the discrete warp, validated against held-out GCPs — not the same points used to fit the warp.
What about the 320× IIRS scale gap?	We never match those GSDs natively; we always match at the coarser common grid and, only if needed, refine downward.
Did you use ISIS/SPICE?	As an optional Tier-1 backend where installed; the default Tier-2/3 path (footprint affine or selenographic rotation) is what runs everywhere, including on the judges' machine.
How generic is this really?	One job schema and one gate table cover all three instruments — branches, not three separate apps.

12 · Risks & Mitigations

Risk	Likelihood	Mitigation
Official TBD dataset is an unfamiliar format	High	Ingest layer is a protocol: if you can build a <code>Pair</code> object, everything downstream runs unchanged; budget hours 0–4 of the finale for a new reader if needed.
No GPU at the venue	Medium	CPU path is mandatory-tested in Phase C: phase-correlation + SIFT/AKAZE + crater graph + phase congruency + MI, LightGlue on small CPU tiles. Dense matcher stays optional.
ISIS/ASP won't install on a team laptop / Windows	High on Windows	Tier 2/3 geometry path is the tested default, not a fallback-in-theory; ISIS is purely a stretch credibility item.
Crater detector fails on smooth mare	High	Hillshade + phase-congruency + LightGlue still run; crater graph is one gate among several, never a single point of failure.
OHRC strip too large for laptop RAM	High	Never load a full strip; tile via COG windows; demo a crop live, show a precomputed full product.
Opposite-azimuth pair still partly fails	Medium	Show SIFT failing worse; show your masked/graceful degrade; the honest failure mask is itself a selling point.
High-latitude IIRS errors (known issue in the literature)	Certain if given polar cubes	Cite the published 73 m/±55° failure point yourself before a judge does; switch CRS to stereographic near the poles.
Overclaiming sub-0.05 px accuracy	Certain if unchecked	Report only held-out RMSE with independent check points, never fit-set residuals.

13 · References

- ISRO / SIH 2026, PS 26166 — official problem statement text.
- ISRO PRADAN / ISSDC MapBrowse — Chandrayaan-2 OHRC, TMC-2, IIRS access.
- ISRO payload notes — OHRC ~0.25 m/3 km, TMC-2 5 m/20 km/±25° stereo, IIRS 0.8–5 μm/~256 bands/80 m/20 km.
- NASA Ames Stereo Pipeline, Chandrayaan-2 cookbook — isisimport, spiceinit, bundle_adjust, mapproject, pc_align, shadow masking.
- USGS ISIS3 — planetary ingest, SPICE, cam2map, control networks.
- SIFT-based registration of Chandrayaan-2 IIRS to LRO-WAC (2025 preprint) — mean RMS ~73 m mid-latitude, failure past ±55°.
- Crater-neighbourhood structure matching for multi-illumination lunar orbiter images (Remote Sensing, 2025).
- Kovesi — Phase congruency: a low-level image invariant.
- Lindenberger, Sarlin, Pollefeys — LightGlue (ICCV 2023).
- Edstedt et al. — RoMa dense matcher.
- COSI-Corr (Leprince / Caltech) — sub-pixel optical correlation.
- OpenCV USAC_MAGSAC — robust model fitting.
- LRO NAC / WAC mosaics, SLDEM2015 / LOLA — geometric control data.